和豆包的对话_0202

用户：

for the exercises, make them 20 questions for both parts

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Understanding Your Request

You'd like to expand the beginner Python exercises (focused on `print()`, `input()`, type casting, and arithmetic) to **20 theoretical questions** and **20 practical coding prompts** (up from 10 each), with no answers, code solutions, or example outputs included. I've kept all exercises strictly aligned with the core topics you specified and ensured they progress gently in difficulty for beginners.

Part 1: 20 Theoretical Exercises (Conceptual Understanding)

These questions test foundational knowledge of `print()`, `input()`, type casting, and arithmetic operations:

1. What data type is returned by the `input()` function by default, regardless of what the user enters?
2. Explain why `print(8 - 3 * 2)` outputs `2` instead of `10`.
3. What is the purpose of type casting in Python, and give one scenario where it's necessary when using `input()` for numbers?
4. What is the difference between `print("4" * 3)` and `print(4 * 3)`?
5. What error occurs if you run `print(int("abc"))`, and why?
6. List the 7 basic arithmetic operators in Python (include `//`, `%`, `**`).
7. How do you print a blank line using the `print()` function?
8. What is the result of `float(10)`? What data type is the output?

9. Explain why `print(9 / 2)` gives a different result than `print(9 // 2)`.
10. How can you print the string `He said, "Hello!"` using `print()` without a syntax error?
11. What is the output of `print(str(7.5) + "5")`, and why?
12. What type cast function would you use to convert the string `"89.9"` to a number that supports- decimal calculations?
13. What does the `%` operator do in Python? Give an example of its use with integers.
14. How do you print the value of a variable `age` (integer) alongside the text `"Your age is: "` in a single- `print()` statement? (Give one method.)
15. What is the result of `15 % 4`, and what does this value represent?
16. What happens if you try to cast a float like `7.9` to an integer using `int()`?
17. What is the purpose of the `**` operator, and what is the output of `print(3 ** 4)`?
18. How do you make `print()` display text without adding a new line at the end?
19. What is the difference between `print(str(5) + str(7))` and `print(5 + 7)`?
20. Why is `num = input("Enter a number: ")` insufficient for arithmetic if `num` is used in `num + 5`? How- do you fix this?

Part 2: 20 Practical Coding Exercises (Hands-On)

These prompts require writing functional Python code (no solutions provided):

1. Print the result of `25 + 18`, `40 - 12`, `15 * 6`, and `72 / 8` — each on a new line with a descriptive label- (e.g., "Division: 9.0").
2. Ask the user to enter two numbers, subtract the smaller from the larger, and print the result (- ensure type casting).
3. Convert a user-input string (e.g., `"123"`) to an integer, then convert that integer to a float, and print- both values with their data types.
4. Calculate the perimeter of a rectangle using `2 × (length + width)` — ask the user for length and- width (as input), then print the perimeter.
5. Ask the user for a number, compute its square (using `**`), and print the result with a label (e.g., "- Square of 5 is 25").
6.

Convert Fahrenheit to Celsius using C = (F - 32) × 5/9 — ask the user for a Fahrenheit value, cast- to float, compute, and print the result.

7. Ask the user for three numbers, calculate their average (sum divided by 3), and print the average.
8. Compute the remainder when a user-input number is divided by 7 (use %), then print a message- like "10 divided by 7 leaves a remainder of 3".
9. Ask the user for their birth year (integer), calculate their current age (subtract birth year from 20- 26), and print the result.
10. Calculate the area of a circle using $\pi \times r^2$ (use $\pi = 3.14159$) — ask the user for radius (float),- compute, and print the area (include m^2 in the output).
11. Ask the user for two numbers, compute their product, then divide the product by 2 — print each- step’ s result (e.g., "5 × 8 = 40; 40 / 2 = 20").
12. Convert a user’ s input (string like "100") to an integer, add 25, then convert the result back to a- string and print the final string value.
13. Calculate the volume of a cube using $side^3$ — ask the user for side length (float), compute, and- print the volume (include m^3 in the output).
14. Ask the user for a number, compute (number + 5) × 3, and print the result with a label.
15. Print the string The total cost is $45.50 using a variable for the cost (float) and an f-string.
16. Ask the user for an integer, compute integer division by 5 (use //), then print the quotient (e.g., "- 17 // 5 = 3").
17. Calculate the sum of the first 10 positive integers (1+2+...+10) using arithmetic, then print the- result with a descriptive message.
18. Ask the user for their height in centimeters (float), convert it to meters (divide by 100), and print- the result (e.g., "180 cm = 1.8 m").
19. What is the output of print(10 - (3 * 2) + 8 / 2)? Write code to verify the result (print only the final- value).
20. Ask the user for two numbers: a base and an exponent. Compute the base raised to the power of- the exponent (use **), then print the result (e.g., "2^5 = 32").

Summary

1. The 20 theoretical questions cover core concepts like data types, operator precedence, type- casting errors, and print()/input() behavior — all aligned with your specified topics.

2. The 20 practical exercises build progressively from basic arithmetic/printing to multi-step-calculations, unit formatting (m^2/m^3), and real-world use cases (area/volume, temperature-conversion).
3. All exercises avoid advanced topics (loops, functions, conditionals) and focus exclusively on `print()`, `input()`, type casting, and arithmetic operations for beginners.

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